

ECE 461 - HOMEWORK 06

$$1) \quad y[m] = x[m] + \frac{1}{4} x[m-1] + \frac{1}{3} x[m] + w[m]$$

with $x[m] = 0$ for $m \neq 1, 2$.

* $x[1], x[2]$ i.i.d. with mean E and variance

* $w[m] \sim \mathcal{N}(0, \sigma^2)$ i.i.d.

1.c)

Recalling that $y[m] = \sum_{k=0}^L h[k] x[m-k] + w[m]$, one finds

$$h[0] = 1, \quad h[1] = \frac{1}{4}, \quad h[2] = 1/3 \quad \text{and} \quad h[m] = 0 \quad \text{for } m \neq 0, 1, 2$$

$L = 3$

$$y[1] = x[1] + w[1]$$

$$y[2] = x[2] + \frac{1}{4} x[1] + w[2]$$

$$y[3] = \frac{1}{4} x[2] + \frac{1}{3} x[1] + w[3]$$

* We estimate y using a matched filter $c[l]$:

$$\hat{y}[m] = \sum_{l=0}^{L-1} c[l] y[m+l]$$

↳ We are looking for $\hat{y}[1] = c[0] y[1] + c[1] y[2] + c[2] y[3]$

* We've derived in class that (ignoring ISI) optimal $c[l]$ which yields the highest SNR is equal to $h[l]$ or in another words coeffs. of $x[1]$ in the equations above

→ Change index $c[0] \rightarrow c_1, c[1] \rightarrow c_2, c[2] \rightarrow c_3$

$$C_1 = 1, \quad C_2 = 1/4, \quad C_3 = 1/3$$

1.b)

$$\hat{y}[1] = 1 \cdot y[1] + \frac{1}{4} y[2] + \frac{1}{3} y[3]$$

$$\begin{aligned} \hat{y}[1] &= (x[1] + w[1]) + \frac{1}{4} (x[2] + \frac{1}{4} x[1] + w[2]) + \frac{1}{3} \left(\frac{1}{4} x[2] + \frac{1}{3} x[1] + w[3] \right) \\ &= \underbrace{\left(1 + \frac{1}{16} + \frac{1}{9} \right) x[1]}_{\text{Signal}} + \underbrace{\left(\frac{1}{4} + \frac{1}{12} \right) x[2]}_{\text{ISI}} + \underbrace{\left(w[1] + \frac{1}{4} w[2] + \frac{1}{3} w[3] \right)}_{\text{Noise}} \\ &= \frac{169}{144} x[1] + \frac{1}{3} x[2] + 1 \cdot w[1] + \frac{1}{4} w[2] + \frac{1}{3} w[3] \end{aligned}$$

$$\text{SINR} = \frac{\mathbb{E}[S_{\text{MF}}^2[m]]}{\mathbb{E}[I_{\text{MF}}^2[m]] + \mathbb{E}[N_{\text{MF}}^2[m]]}$$

$$\mathbb{E}[S_{\text{MF}}^2[m]] = \left(\frac{169}{144} \right)^2 \mathbb{E}[x^2[1]] = \left(\frac{169}{144} \right)^2 E$$

$$\mathbb{E}[I_{\text{MF}}^2[m]] = \left(\frac{1}{3} \right)^2 \mathbb{E}[x^2[2]] = \frac{1}{9} E$$

$$\mathbb{E}\left[\left(w[1] + \frac{1}{4} w[2] + \frac{1}{3} w[3]\right)^2\right] = 1 \mathbb{E}[w^2[1]] + \frac{1}{16} \mathbb{E}[w^2[2]] + \frac{1}{9} \mathbb{E}[w^2[3]]$$

↳ since $w[m]$'s are independent and $\mathbb{E}[w[m]] = 0, \forall m$

$$= \left(1 + \frac{1}{16} + \frac{1}{9} \right) \sigma^2 = \frac{169}{144} \sigma^2$$

$$\text{SINR} = \frac{\left(\frac{169}{144} \right)^2 E}{\frac{1}{9} E + \frac{169}{144} \sigma^2} = \boxed{\frac{\frac{169}{144} \text{SNR}}{\frac{16}{169} \text{SNR} + 1}} = \frac{\left(\frac{13}{12} \right)^2 \text{SNR}}{\left(\frac{4}{13} \right)^2 \text{SNR} + 1}$$

$$2) \quad y[m] = \frac{1}{2} x[m] + \frac{1}{4} x[m-1] + w[m] \quad \text{where}$$

$$\hookrightarrow w[m] \sim \mathcal{N}(0, \sigma^2) \quad \text{i.i.d.}$$

$$\hookrightarrow x[0] = \pm \sqrt{E} \quad \text{with equal prob.}$$

2.a)

We have $y[0] = \frac{1}{2} x[0] + w[0]$ for $x[1] = 0$
 $y[1] = \frac{1}{4} x[0] + w[1]$

→ Similar to last question, coeffs. of matched filter is given by the response function $h[m]$

$$\hat{y}[0] = c_1 y[0] + c_2 y[1] \Rightarrow c_1 = h[0] = \frac{1}{2} \quad \text{and} \quad c_2 = h[1] = \frac{1}{4}$$

$$\begin{aligned} \hat{y}[0] &= \boxed{\frac{1}{2} y[0] + \frac{1}{4} y[1]} = \frac{1}{2} \left(\frac{1}{2} x[0] + w[0] \right) + \frac{1}{4} \left(\frac{1}{4} x[0] + w[1] \right) \\ &= \left(\frac{1}{4} + \frac{1}{16} \right) x[0] + \frac{1}{2} w[0] + \frac{1}{4} w[1] = \underbrace{\frac{5}{16} x[0]}_{\text{Signal}} + \underbrace{\frac{1}{2} w[0] + \frac{1}{4} w[1]}_{\text{noise}} \end{aligned}$$

Note that there is no interference term. ↯

$$\text{Let } X = \frac{5}{16} |x[0]| = \frac{5}{16} \sqrt{E} \quad \text{and} \quad \tilde{W} = \frac{1}{2} w[0] + \frac{1}{4} w[1]$$

$$\tilde{W} \sim \mathcal{N}(\mu_{\tilde{W}}, \sigma_{\tilde{W}}^2) \quad \text{with} \quad \mu_{\tilde{W}} = 0, \quad \sigma_{\tilde{W}}^2 = \left(\frac{1}{2}\right)^2 \sigma^2 + \left(\frac{1}{4}\right)^2 \sigma^2 = \frac{5\sigma^2}{16}$$

$$P(\text{error}) = \frac{1}{2} P(X + \tilde{W} < 0) + \frac{1}{2} P(-X + \tilde{W} > 0)$$

$$= \frac{1}{2} \Phi\left(-\frac{\mu_{\tilde{W}}}{\sigma_{\tilde{W}}}\right) + \frac{1}{2} Q\left(\frac{\mu_{\tilde{W}}}{\sigma_{\tilde{W}}}\right) = Q\left(\frac{\mu_{\tilde{W}}}{\sigma_{\tilde{W}}}\right) = Q\left(\frac{\frac{5\sqrt{E}}{16}}{\sqrt{\frac{5}{16}\sigma^2}}\right) = \boxed{Q\left(\frac{\sqrt{5}}{4} \text{SNR}\right)}$$

2.b) Now since $x[1] \neq 0$, we'll have an interference term.

$$y[0] = \frac{1}{2} x[0] + w[0]$$

$$y[1] = \frac{1}{2} x[1] + \frac{1}{4} x[0] + w[1]$$

→ Matched filter remains the same:

$$\hat{y}[0] = \frac{1}{2} y[0] + \frac{1}{4} y[1]$$

$$= \frac{5}{16} x[0] + \underbrace{\frac{1}{8} x[1]}_{ISI} + \underbrace{\frac{1}{2} w[0] + \frac{1}{4} w[1]}_{\tilde{w}}$$

* We have to consider cases $x[1] = \pm \sqrt{E}$ and find conditional prob. of error

$$P(\text{error}) = P(\text{error} | x[1] = +\sqrt{E}) \cdot \frac{1}{2} + P(\text{error} | x[1] = -\sqrt{E}) \cdot \frac{1}{2}$$

$$= \frac{1}{2} \left[\frac{1}{2} P\left(\underbrace{\frac{5}{16} \sqrt{E} + \frac{1}{8} \sqrt{E} + \tilde{w}}_{x[0] = x[1] = +\sqrt{E}} < 0\right) + \frac{1}{2} P\left(\underbrace{-\frac{5}{16} \sqrt{E} + \frac{1}{8} \sqrt{E} + \tilde{w}}_{x[0] = -\sqrt{E}, x[1] = \sqrt{E}} > 0\right) \right]$$
$$+ \frac{1}{2} \left[\frac{1}{2} P\left(\underbrace{\frac{5}{16} \sqrt{E} - \frac{1}{8} \sqrt{E} + \tilde{w}}_{x[0] = +\sqrt{E}, x[1] = -\sqrt{E}} < 0\right) + \frac{1}{2} P\left(\underbrace{-\frac{5}{16} \sqrt{E} - \frac{1}{8} \sqrt{E} + \tilde{w}}_{x[0] = x[1] = -\sqrt{E}} > 0\right) \right]$$

$$= \frac{1}{2} Q\left(\frac{\frac{7\sqrt{E}}{16}}{\sqrt{\frac{5}{16}\sigma^2}}\right) + \frac{1}{2} Q\left(\frac{\frac{3\sqrt{E}}{16}}{\sqrt{\frac{5}{16}\sigma^2}}\right) = \frac{1}{2} Q\left(\frac{7\sqrt{E}}{4\sqrt{5}\sigma}\right) + \frac{1}{2} Q\left(\frac{3\sqrt{E}}{4\sqrt{5}\sigma}\right)$$

$$P(\text{error}) = \frac{1}{2} \left[Q\left(\frac{7}{4\sqrt{5}} \text{SNR}\right) + Q\left(\frac{3}{4\sqrt{5}} \text{SNR}\right) \right]$$

c) As $\text{SNR} \rightarrow \infty$, Q functions will go to zero and thus $P_e \rightarrow 0$ at high SNR regime.

↳ But note that this is strictly due to the fact that coefficient of $x[0]$ (signal) is larger than $x[1]$ (interference) in the matched filter:

$$\hat{y}[0] = \frac{5}{16} x[0] + \frac{1}{8} x[1] + \tilde{w} \Rightarrow \frac{5}{16} > \frac{1}{8}$$

→ Thus increasing SNR, amplifies to (signal - interference)/noise ratio which decreases error prob.

If there were more interference than signal, increasing SNR would degrade system performance

3) $y[m] = x[m] + 2x[m-1] + w[m]$ where:

$\rightarrow x[m] = 0$ for $m \neq 1, 2$.

$\rightarrow x[1], x[2]$ i.i.d. with mean E and variance

$\rightarrow w[m] \sim \mathcal{N}(0, \sigma^2)$ i.i.d.

$$y[1] = x[1] + w[1]$$

and
$$h[m] = \begin{cases} 1, & m=0 \\ 2, & m=1 \\ 0, & \text{else} \end{cases}$$

$$y[2] = x[2] + 2x[1] + w[2]$$

We wish to design an equalizer of the form: $\hat{y}[1] = c_1 y[1] + c_2 y[2]$

* To minimize SINR, we again choose the matched filter coeffs. equal to $h[m]$:

$$c_1 = h[0] = 1, \quad c_2 = h[1] = 2$$

$$\begin{aligned} \hat{y}[1] &= y[1] + 2y[2] = (x[1] + w[1]) + 2(x[2] + 2x[1] + w[2]) \\ &= \underbrace{5x[1]}_{S_{MF}[1]} + \underbrace{2x[2]}_{I_{MF}[1]} + \underbrace{w[1] + 2w[2]}_{N_{MF}[1]} \end{aligned}$$

$$\text{SINR} = \frac{\mathbb{E}[S_{MF}^2[m]]}{\mathbb{E}[I_{MF}^2[m]] + \mathbb{E}[N_{MF}^2[m]]} = \frac{25E}{4E + (1^2 + 2^2)\sigma^2} = \boxed{\frac{25 \text{ SNR}}{4 \text{ SNR} + 5}}$$